V S Susi Krishna

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.01
ADITYA VIDYASHRAM PORAIYUR VILLIANUR PUDUCHERRY	Class XII (CBSE)	2023	95.00%
ACHARIYA BALA SIKSHA MANDIR THENGAITHITTU PONDY	Class X (CBSE)	2021	94.20%

WORK EXPERIENCE

Software Engineering Intern | *Google* | *Bangalore*

- Contributed to the Google Play Store Android backend by enhancing existing production code using **Java** and **Kotlin**.
- Migrated a production feature from a legacy API to a newer API, reducing request latency while improving maintainability.
- Developed and executed comprehensive testing, including **GenAI-assisted Android emulator validation**, ensuring feature correctness and reliability before production deployment.

PROJECTS

Delay-Compensated Interval Asynchronous SGD | *Mini Project* | *Mentor: Prof. Sathya Peri*

- Built a **128-thread C++ deep learning training engine** using OpenMP and MiniDNN, supporting Sequential, Hogwild!, and Interval-Asynchronous execution modes alongside custom optimization schemes to maximize CPU parallelism.
- Unified **DC-ASGD** compensation with **Interval-Asynchrony** via Taylor-based Hessian approximations to stabilize updates.
- Designed an adaptive parameter probing algorithm using **binary search** and **neighborhood fine-tuning** to discover optimal thread counts and interval sizes, eliminating manual tuning to drive a **32% speedup on CIFAR-10**.

Linux Network Fairness | *Course Project* | *Github*

- Developed an **eBPF-based network traffic control system** to enforce fair bandwidth allocation across Linux cgroups.
- Implemented **eBPF** programs at **TC** and **XDP** hooks for packet classification and per-cgroup rate limiting.
- Built user-space controllers to manage **BPF maps** and dynamically enforce bandwidth allocation policies under changing network contention, enabling fair scheduling with minimal packet-processing overhead before the Linux **qdisc** layer.

Disha: Role and Permission Management System | *Course Project* | *Github*

- Built a distributed **Authorization System** for microfrontend-microservice stacks using **Node.js, React, and MongoDB**.
- Implemented **frontend-to-backend sandboxing** by mapping microfrontends to microservices and using **microservice:api:action** permission keys to enforce fine-grained API authorization and isolate network requests.
- Developed a modular **RBAC and ABAC framework** utilizing Rule Groups, Roles, and temporary permissions to enforce context-aware, time-bound, and least-privilege access control across all distributed microservices

Hybrid Quantum-Classical Compiler | *Course Project*

- Architected a custom lexer and parser for a **hybrid quantum-classical** programming language in **C++**, seamlessly unifying quantum gate operations, qubit measurements, and classical control flow within a single syntax stream.
- Designed a formal context-free grammar and **AST** to enable downstream semantic analysis and quantum IR generation.

Memory Management in GemOS | *Course Project* | *Github*

- Implemented custom **mmap, munmap, and mprotect syscalls** in gemOS to manage VMAs, handling merging and splitting.
- Handled **lazy page faults** via four-level page tables and built a **Copy-on-Write fork** featuring custom handle_cow_fault logic to optimize memory usage during process creation and manage write-protected shared pages.

SKILLS

Programming Languages: C, C++, Java, Kotlin, Go, Python, JavaScript, HTML/CSS, RISC-V Assembly

Frameworks & Technologies: React, Node.js, Android, Linux, Git, MySQL, MongoDB, eBPF, Flutter

Competitive Programming: Codeforces -- **Max Rating: 1504 (Specialist)**

RELEVANT COURSES

Core Courses: Algorithms, Data Structures, Operating Systems, Computer Networks, Database Management Systems, Computer Architecture, Theory of Computation, Compilers, Software Engineering, Parallel & Concurrent Programming

Math & ML Courses: Discrete Maths, Foundations of Machine Learning, Deep Learning, Security in Intelligent System, Optimization, Linear Algebra, Probability & Random Variables, Computational Number Theory & Algebra, Calculus

SCHOLASTIC ACHIEVEMENTS

- **NSEC :** State Topper in National Standard Examination in Chemistry (NSEC).

POSITIONS OF RESPONSIBILITY

Core Member, **Lambda - Software Development Club**, IIT Hyderabad

Developed the **Go** backend for the **IITH Dashboard** and the **Flutter** frontend for the **KRC Library** application.

Sunshine UG Mentor, IIT Hyderabad

Mentored **20+** first-year students and organized orientation activities for **100+** students.

Core Member, **Kludge - Cybersecurity Club**, IIT Hyderabad

Created CTF challenges and conducted introductory cybersecurity workshops and hands-on learning sessions.

EXTRACURRICULAR

Student Volunteer at National Service Scheme (NSS)